Gulf Coastal Resilience Tool User Guide

Building Resilience in the Gulf through Nature-based Solutions

Nature-based solutions (NBS) are projects that support or mimic nature to reduce risk and provide benefits to people and nature. The Nature Conservancy's Gulf Coastal Resilience tool features "NBS Opportunity Maps" that show the potential for implementing nature-based solutions across the coastal watershed counties of the Gulf. These Opportunity Maps display where an NBS practice is geographically suitable and benefits both people and nature. These benefits include areas of habitat priority (SECAS) and flood risk (NOAA). If you are interested in learning more about planning and building Nature-based Solutions, please visit www.nature.org/SUNS.

For information about this web application or related data, please contact:

Christine Shepard at cshepard@TNC.ORG

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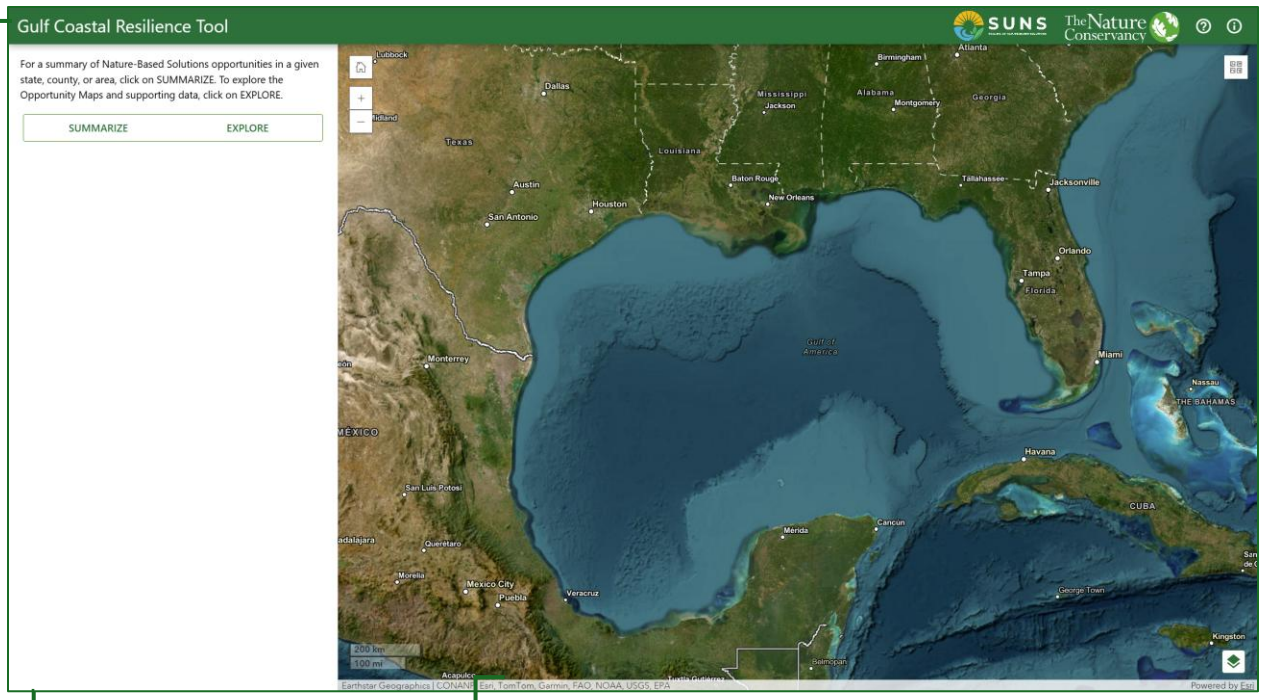
Layout and Content



App Panel

Map

Header

Summarize Panel

State and County Selection

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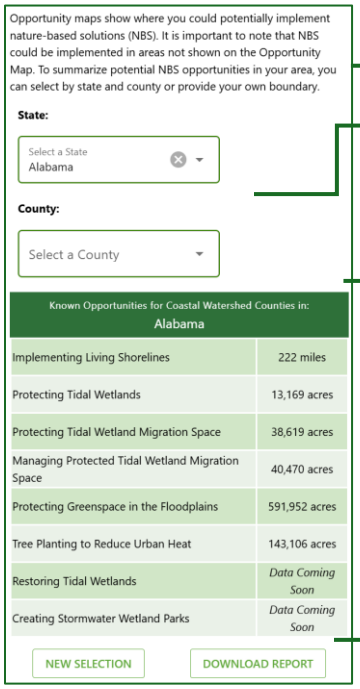
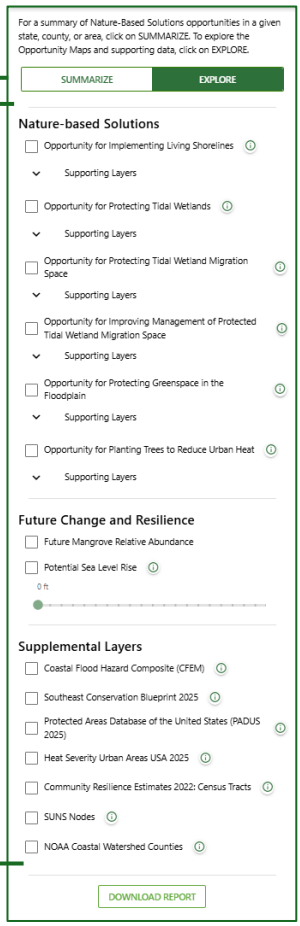
Map Layers

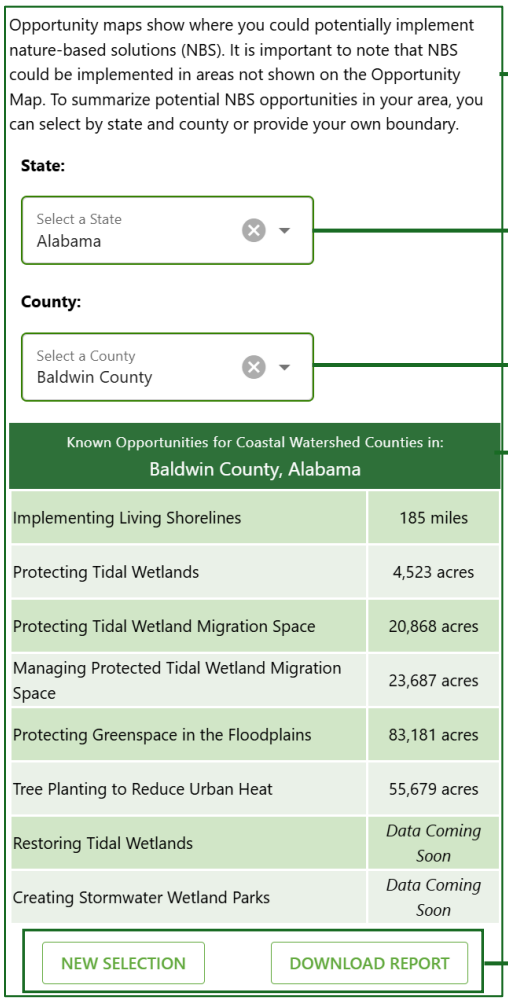




Header

	SUNS Logo – Links to the SUNS website
	TNC Logo – Links to TNC SUNS website
	View application user guide dialog
	View information dialog about this application

Summarize



Summarize panel information

Dropdown component used to select a State (required)

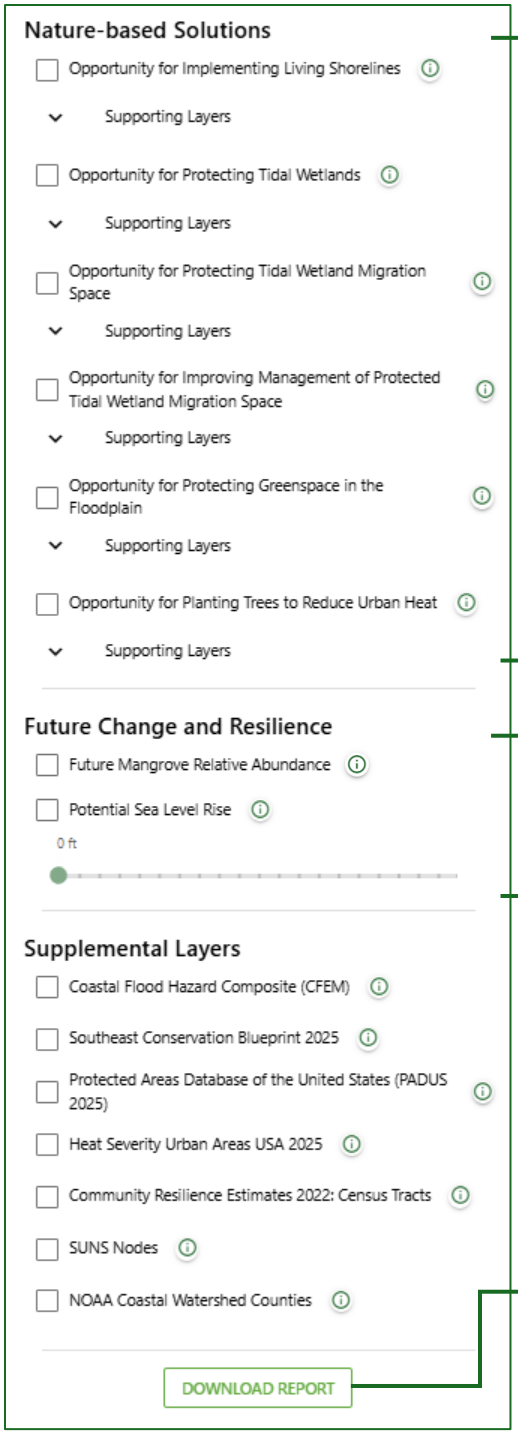
Dropdown component used to select a County (optional)

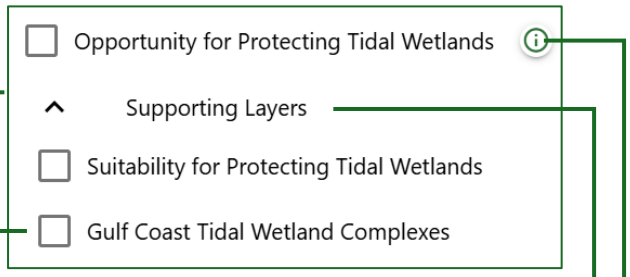
Results table for selected state or county

New Selection: Reset state and county selections

Download Report: Automatically downloads a PDF report containing selections, results, a map screenshot, and extra resources

Explore

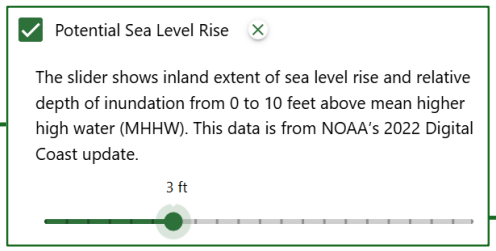




Expand layer's supporting layers

Show/hide layer info

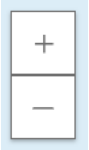
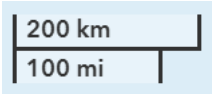
Turn on/off layer



Adjustable slider used to display specified sea level

Download Report: Automatically downloads a PDF report containing a map screenshot and extra resources

Map Tools and Widgets

	Zoom to default extent
	Zoom in and zoom out
	Upload zipped shapefile and add it to the map (Explore only)
	Draw an area of interest on the map (Explore only)
	Basemap selector
	Display or hide draggable legend
	Map scale

Area of Interest

1. **Go to the Explore tab**

   You must start from the Explore tab to upload a shapefile or draw an area of interest.

2. **Click on the upload or draw button**

   Decide whether you want to upload an existing shapefile or manually draw your area of interest.

3. **If using Upload, select a <u>zipped</u> shapefile**
   Uploaded files must be zipped shapefiles; once selected, they are added to the map automatically.

   - After clicking **Upload**, your file explorer will open.
   - Navigate to and select a **zipped shapefile** (.zip containing the shapefile components).
   - Confirm your selection; the file is automatically added to the map.

4. **If using Draw, create your polygon**

   Use the draw tools to outline your area of interest directly on the map.

   - After clicking **Draw**, the draw tools will appear on the map.

   - Click the button to begin drawing.
   - Click on the map to add vertices and outline your shape.
   - When finished, **double click** the final vertex and then click **outside** of your shape to generate results.

5. View area of interest results

Once your area is defined, the app displays the results in the right-side panel.

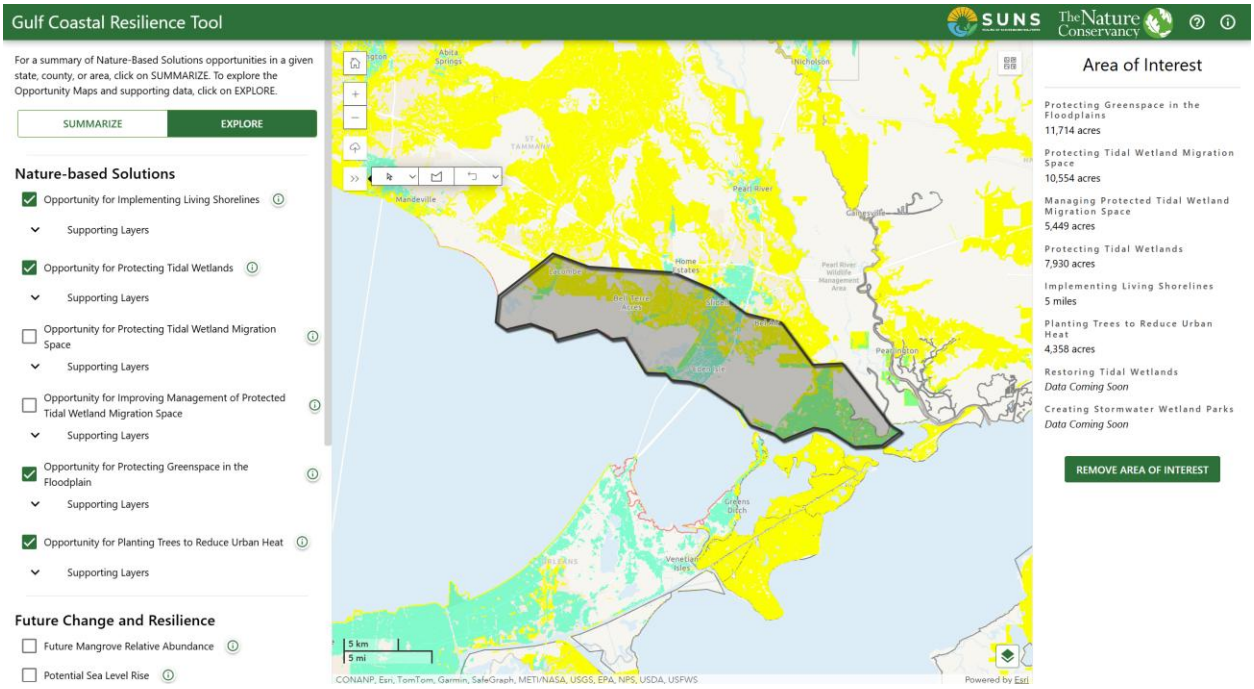


Figure 1: Example of drawn area of interest with NBS layers turned on

Additional Resources

Opportunity for Protecting Tidal Wetlands

Tidally influenced wetlands, such as salt marshes, mangroves, brackish marshes, and tidal flats can be legally protected through outright acquisition or conservation easements. This map displays opportunities for protecting wetlands that are good opportunities based on their resilience. Wetlands with higher resilience are those that have better ability to migrate inland to adjacent lowlands with sea level rise.

Opportunity for Managing Protected Tidal Wetland Migration Space

While some tidal wetland migration space is currently under a protected status, it is unclear if land managers know that the land is migration space or if they are actively managing the land to facilitate the inland migration of tidal wetlands. This map displays opportunities for improving management of protected wetland migration space through outreach and education to land managers and implementation of best management practices for facilitating wetland migration.

Opportunity for Protecting Greenspace in the Floodplain

Strategic land protection, where specific undeveloped open spaces in the floodplain are protected in perpetuity, is a viable nonstructural solution for reducing flood risk and protecting fish and wildlife habitat in the floodplain.

Opportunity for Protecting Tidal Wetland Migration Space

Tidal wetland migration space is the area of low-lying land adjacent to tidal wetlands that is potentially suitable for supporting tidal habitats under future sea level rise scenarios. This map displays opportunities for protecting wetland migration space behind resilient wetlands through acquisition or conservation easement.

Opportunity for Implementing Living Shorelines

Living shorelines are protected, stabilized coastal edges made of natural materials such as plants, natural materials such as plants, rock, or beneficial use of dredged material. These natural materials can be placed on their own or in combination with articulated blocks,

riprap, or breakwaters. This map displays where implementing living shorelines may or may not be suitable based on NOAA's Living Shoreline Suitability Models and TXGLO's Living Shoreline Suitability Model. Areas where these models have not been performed show as No Data Available.

Opportunity for Tree Planting to Reduce Urban Heat

Urban heat islands form when natural landscapes are replaced with dense infrastructure that traps heat. Without sufficient tree canopy or green space to provide cooling, temperatures in some neighborhoods can be several degrees higher than in greener parts of the same city. This map displays opportunities where planting trees to increase tree canopy could reduce urban heat severity in these communities.

PDF Report

Monday, April 20, 2026

Scaling up Nature-based Solutions

Protecting Greenspace in the Floodplains

Protecting Tidal Wetland Migration Space

Managing Protected Tidal Wetland Migration Space

Protecting Tidal Wetlands

Implementing Living Shorelines

Opportunity for Planting Trees to Reduce Urban Heat

Known NBS Opportunities	Mobile County, Alabama
Protecting Greenspace in the Floodplains	82902 acres
Protecting Tidal Wetland Migration Space	17751 acres
Managing Protected Tidal Wetland Migration Space	16783 acres
Protecting Tidal Wetlands	8646 acres
Implementing Living Shorelines	37 miles
Planting Trees to Reduce Urban Heat	59379 acres
Restoring Tidal Wetlands	0 acres
Creating Stormwater Wetland Parks	0 acres

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Monday, April 20, 2026

Additional Resources

Nature-based solutions (NBS) can be used to reduce risk and benefit people and nature along the Gulf coast. TNC has created "Opportunity Maps" that show the potential for implementing nature-based solutions across the coastal watershed counties of the Gulf. These Opportunity Maps display where an NBS practice is geographically suitable and benefits both people and nature. These benefits include areas of habitat priority (SECAS) and flood risk (NOAA).

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Date of report creation

You can update the screenshot anytime by adjusting the map (zoom, pan, or change layers) and re-download. The image always reflects whatever is visible on the map at that moment.

Map legend

Selection results table

Additional information about each opportunity map layer